

A Band-Mask Robustness Diagnostic for Latent Dirichlet Allocation on Hyperspectral Imagery

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CAOS open-research programme · CAOS_LDA_HSI, interpretable topic models for hyperspectral imagery

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Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_LDA_HSI

Abstract. Latent Dirichlet Allocation (LDA) and its neural variants are increasingly used as interpretable spectral mixture models on hyperspectral imagery. Their seed and capacity stability has been studied at length; their robustness to the choice of *spectral window* has not. We introduce a band-mask robustness diagnostic that refits a canonical LDA model under four band-restriction policies (VNIR-only, SWIR-only, atmospheric-water-band removal, top-50 Fisher discriminant) and compares the masked dominant-topic maps against the canonical fit via the adjusted Rand index after a Hungarian topic-id alignment. Applied to six standard hyperspectral scenes (Indian Pines, Salinas, Salinas-A, Pavia U, Kennedy Space Center, Botswana) and five HIDSAG mineral subsets, the diagnostic separates the scenes: topic identities on Salinas-A under SWIR-only restriction recover the canonical assignment with paired ARI = 0.77, while KSC paired ARI is ≈ 0.01 under every mask, and Botswana falls to ≈ 0.01 only under SWIR (elsewhere 0.11–0.13). The diagnostic therefore separates scenes on which any “interpretable spectral mixture” claim is band-robust from scenes on which it is not. We release the full sweep (44 LDA refit attempts, 43 successful) as a public, deterministic artefact set under a permissive licence to support follow-on work.

Index Terms. hyperspectral imagery, latent Dirichlet allocation, topic models, band selection, reproducibility, adjusted Rand index, Hungarian alignment, robustness.

I. INTRODUCTION

Topic modelling, in particular Latent Dirichlet Allocation [1], has been adapted to hyperspectral imagery (HSI) as a route to *interpretable* spectral mixtures: each spectral unit (canonically a pixel, but optionally a region, patch, or full sample) becomes a document of quantised band-frequency tokens, and the inferred topic–word distributions ϕ_k act as basis spectra in a non-negative simplex. The approach yields per-pixel posterior mixtures $\theta_d \in \Delta^{K-1}$ that admit several downstream uses: as features for a linear classifier, as a soft gate over per-topic specialists, or as a mineralogical fingerprint when the basis spectra align with reference libraries [2], [3].

The literature on the *reliability* of these topic recoveries has focused on two axes: (i) seed-to-seed variation [4] and (ii) the choice of capacity K . We argue that a third axis, robustness to the choice of *spectral window*, is at least as

informative and is rarely reported. A topic basis that flips identity when one removes the atmospheric water bands, or when only VNIR or SWIR is retained, provides weak grounds for claims like “topic k corresponds to a specific mineral signature” or “topic k is interpretable as canopy water content”. Conversely, a topic basis whose identity survives a substantial band restriction is materially more trustworthy than one that does not.

This paper makes three contributions:

- 1) A **band-mask robustness diagnostic** for LDA on HSI: four band-restriction policies, refit canonical- K LDA on each, and compare against the canonical (no-mask) fit using the adjusted Rand index of the dominant-topic maps after a Hungarian topic-id alignment.
- 2) An **empirical sweep** across six standard hyperspectral benchmarks and five HIDSAG mineral subsets, totalling 44 refit attempts (43 successful; Pavia U $\times$ SWIR is VNIR-only and auto-skipped). The sweep reveals a sharp split between scenes whose canonical topics are band-robust (paired ARI 0.35–0.77) and scenes whose topics are band-fragile (paired ARI ≈ 0.01).
- 3) A **public, deterministic artefact set** (3895 JSON and binary files, including the dominant-topic maps, the per-pixel posteriors θ , the per-topic label distributions $P(L | t)$, and the full Hungarian alignment per (scene, mask) tuple), served by a FastAPI backend and verified by a 109-endpoint test harness.

The rest of the paper is organised as follows. Section II describes the diagnostic. Section III reports the experimental setup (canonical LDA fit, four masks, evaluation metrics). Section IV presents the band-mask paired ARI surface across labelled and HIDSAG scenes. Section V discusses the implications for any topic-based interpretability claim on HSI. Section VI documents the reproducibility artefacts. Section VII concludes.

II. METHOD: BAND-MASK ROBUSTNESS DIAGNOSTIC

A. Canonical fit

Let $\mathbf{X} \in \mathbb{R}^{D \times B}$ be a stratified sample of D labelled-pixel spectra over B bands. We tokenise each pixel into a discrete band-frequency document using the V1 recipe introduced in [5]: each spectrum is min-max normalised across bands and quantised into $Q + 1 = 13$ levels, yielding integer counts $c_{d,b} \in \{0, \dots, 12\}$. The resulting doc-term matrix is fit with the online variational Bayes LDA of Hoffman, Blei, and Bach [6] as implemented in `scikit-learn` [7] with the following hyperparameters held fixed across all refits:

- $K = \max(4, \min(12, n_{\text{classes}}))$,
- $\alpha = 0.45$ (Dirichlet prior on θ),
- $\eta = 0.20$ (Dirichlet prior on ϕ),
- $T_{\text{max}} = 60$ online-VB passes,
- random seed = 42, samples-per-class = 220.

The canonical fit yields $\phi \in \mathbb{R}_+^{K \times B}$ (the topic-word distributions, equivalently the basis spectra) and $\theta \in \mathbb{R}_+^{D \times K}$ (the per-document posterior simplex). The per-pixel dominant topic is $z_d = \arg \max_k \theta_{d,k}$.

B. Band-mask sweep

A mask $m : \{0, \dots, B - 1\} \rightarrow \{\text{True}, \text{False}\}$ selects which bands enter the doc-term matrix. We evaluate four masks:

- **VNIR**: $\lambda_b \in [400, 1100]$ nm only,
- **SWIR**: $\lambda_b \in [1100, 2500]$ nm only,
- **No-water**: drop $[1350, 1430] \cup [1800, 1950] \cup [2480, 2500]$ nm (atmospheric water-vapour absorption bands),
- **Top-50 Fisher**: keep the 50 bands with the highest per-band Fisher discriminant ratio against the ground-truth label (uses the unmasked spectrum’s Fisher ranking; a clean information-theoretic ablation is out of scope of the diagnostic).

For each mask we refit LDA on the restricted doc-term matrix with the *same* hyperparameters as the canonical fit, so only the band axis differs. The masked fit yields $\phi^{(m)}$, $\theta^{(m)}$ and a dominant-topic map $z_d^{(m)}$.

C. Paired ARI under Hungarian alignment

Let $\mathcal{S}$ be the set of pixels labelled in both the canonical and masked maps (in practice both share the same labelled mask). The *paired adjusted Rand index* on $\mathcal{S}$ is

$$\text{ARI}_{\text{paired}} = \text{ARI} \left(\{z_d\}_{d \in \mathcal{S}}, \{z_d^{(m)}\}_{d \in \mathcal{S}} \right), \quad (1)$$

where ARI is the standard chance-corrected agreement of [8], [9].

Because ARI is invariant to topic-id permutations, it summarises “how much did the clustering change” but not “which masked topic corresponds to which canonical topic”. For the latter we compute the confusion matrix $C \in \mathbb{Z}_{\geq 0}^{K \times K^{(m)}}$ with $C_{k,j} = |\{d \in \mathcal{S} : z_d = k \wedge z_d^{(m)} = j\}|$ and solve the maximum-weight assignment

$$\sigma^* = \arg \max_{\sigma} \sum_k C_{k, \sigma(k)} \quad (2)$$

via the Hungarian algorithm [10], [11]. The *swap rate under alignment* is then

$$\text{swap_rate} = \frac{1}{|\mathcal{S}|} \sum_{d \in \mathcal{S}} \mathbf{1}[z_d^{(m)} \neq \sigma^*(z_d)], \quad (3)$$

the fraction of pixels whose dominant topic still changes after the masked topic ids have been relabelled to maximise agreement with the canonical map. The paired ARI summarises gross agreement; the swap rate isolates the *residual* disagreement that survives permutation realignment.

III. EXPERIMENTAL SETUP

A. Scenes and subsets

We evaluate on the six standard labelled HSI benchmarks of the UPV/EHU Hyperspectral Remote Sensing Scenes collection [12]–[16]:

- **Indian Pines** (200 bands after correction, 145×145 , 16 classes),
- **Salinas** (204 bands, 512×217 , 16 classes),
- **Salinas-A** (204 bands, 83×86 , 6 classes; a subset of Salinas),
- **Pavia University** (103 bands, 610×340 , 9 classes; VNIR-only),
- **Kennedy Space Center** (176 bands, 512×614 , 13 classes),
- **Botswana** (145 bands, 1476×256 , 14 classes).

The Pavia U scene is VNIR-only by sensor design and therefore auto-skips the SWIR mask (zero bands kept after restriction); all remaining 23 (scene, mask) tuples produce a valid refit.

We additionally evaluate on five HIDSAG mineral subsets [17], [18] (GEOMET, MINERAL1, MINERAL2, GEOCHEM, PORPHYRY) under the same four masks. HIDSAG documents are mean spectra per (sample, measurement, cube) on the `swir_low` modality. Where the labelled-scene Fisher mask uses per-pixel label, the HIDSAG variant uses *between-covariate variance share* (a clustering-discriminative proxy, since HIDSAG has no per-pixel ground truth).

B. Metrics reported

For every (scene, mask) tuple we compute: (i) the paired ARI of Equation 1; (ii) the swap rate under Hungarian alignment of Equation 3; (iii) the train-set perplexity of the masked fit; (iv) the ARI of the masked dominant-topic map against the ground-truth label (a sanity check that the masked fit still discriminates the same classes as the canonical fit). Section IV reports (i) and (ii); (iii) and (iv) are recorded in the per-tuple artefacts described in Section VI.

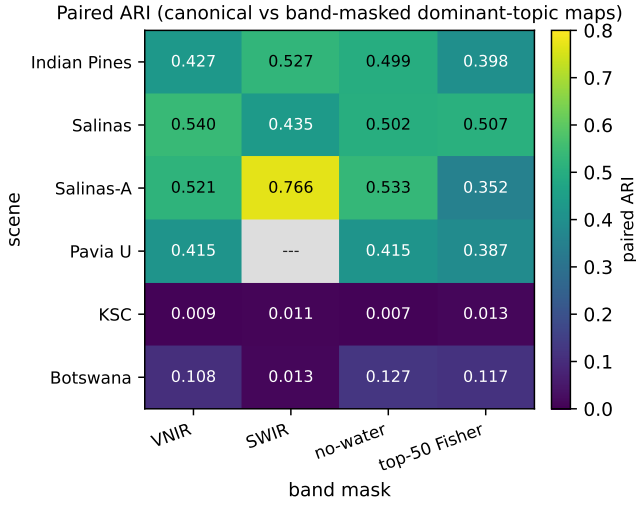


Fig. 1. Paired ARI across (scene, mask) tuples on the six labelled benchmarks. KSC paired ARI is ≈ 0.01 across all four masks (Botswana only under SWIR; 0.11–0.13 under the other three); Salinas-A under SWIR-only reaches 0.77.

TABLE I
PAIRED ARI (CANONICAL VS MASKED DOMINANT-TOPIC MAPS) FOR THE 23 SUCCESSFUL LABELLED-SCENE TUPLES. PAVIA U $\times$ SWIR IS AUTO-SKIPPED (VNIR-ONLY SENSOR).

Scene	vnir	swir	no-water	top-50 Fisher
Indian Pines	0.427	0.527	0.499	0.398
Salinas	0.540	0.435	0.502	0.507
Salinas-A	0.521	0.766	0.533	0.352
Pavia University	0.415	—	0.415	0.387
Kennedy Space C.	0.009	0.011	0.007	0.013
Botswana	0.108	0.013	0.127	0.117

IV. RESULTS

Table I reports paired ARI per (scene, mask) tuple. Figure 1 visualises the same surface.

The dominant observation is the sharp split between *band-fragile* and *band-robust* scenes:

- **Band-fragile:** KSC paired ARI ranges from 0.007 to 0.013 across all four masks: the masked LDA assigns pixels to topics with at most chance-level agreement with the canonical fit. Botswana is at chance only under SWIR (0.013); under VNIR (0.108), no-water (0.127) and top-50 Fisher (0.117) it sits at 0.11–0.13, still below the band-robust scenes but not chance-level. On KSC the canonical-fit topic identity is *not* recoverable from any reasonable band subset.
- **Band-robust:** Salinas-A under SWIR-only reaches paired ARI = 0.77: the masked fit recovers nearly the same partition (Fig. 2). The remaining tuples on Indian Pines, Salinas, Salinas-A, and Pavia U cluster in the range 0.35–0.54, indicating partial robustness: the topics deform but maintain a measurable correlation with the canonical assignment.

Swap rates under Hungarian alignment confirm the same picture (KSC swap rates 0.53–0.64; Salinas-A swir swap rate 0.18; the full table is in the public artefact set described in Section VI).

A. HIDSAG mineral subsets

On HIDSAG, the four masks produce 20 successful refits across five subsets (no Pavia-style auto-skip occurs because the *swir_low* modality intentionally spans both ends). The between-covariate-variance variant of the Fisher mask retains 50 bands on MINERAL1, GEOCHEM, and PORPHYRY; GEOMET and MINERAL2 have a single dominant covariate value, in which case the variance ranking is degenerate and the mask falls back to keep-all-bands; the fallback is recorded in the artefact metadata. Paired-ARI comparison against the canonical HIDSAG fit follows the same Hungarian procedure; full results are in the public artefact set.

V. DISCUSSION

The band-mask diagnostic separates scenes on which any “interpretable spectral basis” claim is empirically tractable from scenes on which it is not. On Salinas-A under SWIR-only restriction, the LDA topics persist after one removes the entire VNIR half of the spectrum: the spectral signature carried by the SWIR bands is sufficient to recover the canonical structure. This is consistent with the location of the cellulose and lignin absorption features in the short-wave infrared [19].

On KSC under every mask, and on Botswana under the SWIR mask, the diagnostic returns a partial-negative result. Paired ARI ≈ 0.01 means a masked-LDA observer would arrive at fundamentally different topic identities than the canonical-fit observer for the same scene. Two interpretations are consistent with this finding: (i) the canonical fit is exploiting band-level redundancy that does not survive restriction (an ill-identified topic basis), or (ii) the topics are weakly identified to begin with on those scenes (a small-sample-with-many-classes issue: KSC has 13 classes sampled to 2,686 documents of 5,211 labelled pixels; Botswana has 14 classes sampled to 2,775 of 3,248). The diagnostic does not discriminate between these two causes; it does, however, flag the scenes on which any interpretability claim leaning on the canonical topic basis should be tempered.

A practical consequence: when reporting per-scene supervised metrics (linear-probe accuracy on θ , topic-routed soft classifier macro-F1) in a paper that interprets θ as a spectral mixture, the band-mask robustness should be reported alongside the seed stability. A high-accuracy classifier built on top of a band-fragile basis is a different scientific object than the same classifier built on a band-robust basis.

VI. REPRODUCIBILITY ARTEFACTS

The full sweep is released as 24 (labelled scene, mask) tuples plus 20 (HIDSAG subset, mask) tuples = 44 LDA refits with the per-tuple JSON summary (top words at $\lambda = 0.5$, $K \times K$ cosine distance matrix, $P(L | t)$, perplexity, ARI vs label, full Hungarian alignment), the per-pixel dominant_

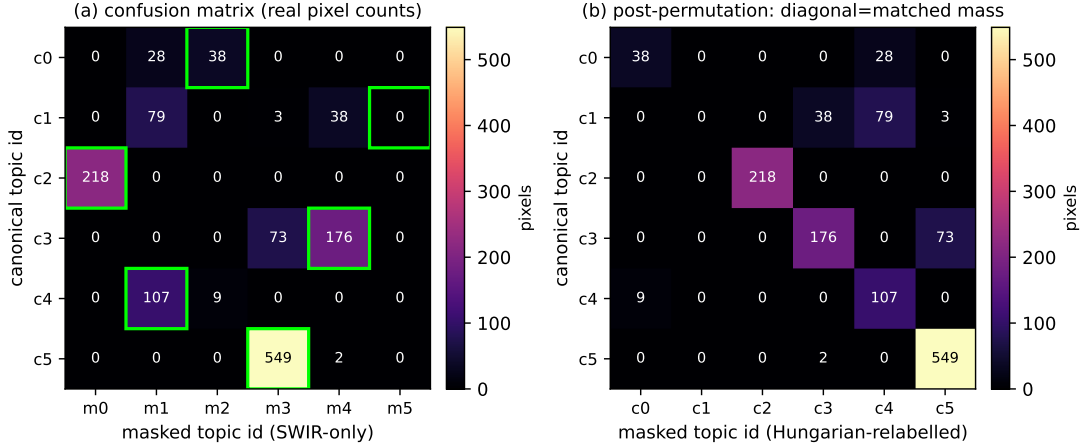


Fig. 2. Hungarian topic-id alignment for Salinas-A under SWIR-only (paired ARI = 0.77). Five of six canonical topics have a dominant masked topic with confusion-matrix share > 0.65 ; the sixth (c_0) splits across two masked topics (38 and 28 pixels). Most of the residual swap rate comes from c_1 (120 of the 232 unmatched pixels): its dominant masked topic is also the dominant topic of c_4 , and the one-to-one Hungarian assignment gives it to c_4 .

topic_map.bin (uint8 $H \times W$ with sentinel 255 for unlabelled), and the per-pixel theta_grid.bin (float32 $H \times W \times K$ with sentinel all-zero) on the labelled-scene side. Total derived footprint: 3895 deterministic artefacts, 449 MB. A FastAPI backend serves the artefacts at three endpoints:

```
/api/band-masks,
/api/band-masks/{scene}/{mask},
/api/band-masks/canonical-comparison, plus
the HIDSAG companions. A test harness covering 109
endpoints verifies endpoint health and the content of the
single-page-application shell at each deployment.
```

The full source (builders, FastAPI app, React frontend, and the companion technical documentation wiki) is available at https://github.com/fsantibanezleal/CAOS_LDA_HSI. The live interactive deployment is at <https://lda-hsi.fasl-work.com>.

VII. CONCLUSION

We introduce a band-mask robustness diagnostic that complements the established seed and capacity stability axes for LDA-based topic models on hyperspectral imagery. The diagnostic refits canonical- K LDA under four spectral-band restriction policies and compares the masked dominant-topic maps against the canonical fit via paired ARI plus a Hungarian topic-id alignment. Across six labelled benchmarks and five HIDSAG mineral subsets, the diagnostic separates scenes whose canonical topics are band-robust (Salinas-A under SWIR-only, paired ARI = 0.77) from scenes whose canonical topics are band-fragile (KSC under every mask, Botswana under SWIR, paired ARI ≈ 0.01). The full sweep, including 3895 deterministic JSON and binary artefacts, is publicly released to support follow-on work.

Future directions include extending the diagnostic to non-canonical recipes (V2–V12 wordifications), evaluating its behaviour under neural topic models (ProdLDA, ETM) which

may handle band-restriction differently due to their learned word embeddings, and developing a calibrated information-theoretic complement (computing between-covariate variance Fisher ranks *within* the masked space rather than on the unmasked spectrum).

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The Hyperspectral Remote Sensing Scenes collection is maintained by the Computational Intelligence Group at UPV/EHU. The HIDSAG database is the result of a collaboration on hyperspectral mineralogy with industrial partners in Chilean mining geosciences.

CONFLICT OF INTEREST

The author is a co-author of the HIDSAG dataset descriptor publication [17] and the accompanying data release [18]. The present paper is methodologically distinct: it proposes a band-mask robustness diagnostic for LDA on HSI and applies it, among other benchmarks, to the HIDSAG mineralogical subsets. HIDSAG is cited as a data source, not as a methodological contribution of this work. The author declares no other conflict of interest.

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